sg13g2_stdcell_hv — Generation of a Thick-Oxide (3.3 V) Standard Cell Library

IHP SG13G2 open PDK · derivation, all views, verification and physical sign-off

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Scope and result

The IHP SG13G2 open PDK ships no 3.3 V standard cells: its thick-oxide devices appear only in the `sg13g2_io` pad ring. `sg13g2_stdcell_hv` fills that gap — **all 84 cells** of `sg13g2_stdcell` rebuilt on `sg13_hv_nmos` / `sg13_hv_pmos`, same topology, pin names and pin order, named `sg13g2_hv_*` so both libraries coexist in one netlist.

Deliverable	Coverage	Status
SPICE netlist (`spice/`)	84 cells, 924 devices	verified against 3 independent views
CDL netlist (`cdl/`)	84 cells + 2 tie cells	LVS reference, all 68 drawn cells match
Verilog (`verilog/`)	84 modules	shared `ihp_*` UDPs, deliberately not duplicated
xschem symbols / schematics	84 + gallery sheet	netlist-equivalence proven
GDS layout (`gds/`)	68 cells (66 retargeted + 2 tie cells built here)	**DRC clean, LVS clean**
LEF abstracts (`lef/`)	68 macros + `CoreSiteHV` site	generated from the GDS, pin sets verified against CDL
Liberty NLDM (`lib/`)	66 cells, 600 timing tables	combinational + sequential; areas from layout

Physical sign-off, one fixed invocation of the PDK's own klayout decks — on the abutted two-row array and on the stricter **shared-rail array** (rows at the true 7.14 µm pitch, mixed and mirrored vertical neighbours — what a placed block actually produces):

Check	This library	IHP thin-oxide control (same harness)
DRC, cell rules	**0**	0
DRC, cell rules, shared-rail mixed array	**0**	—
DRC, chip-level metal density	7	6
LVS	**68** / **68 match**	fill cells fail identically without the documented relaxation

The density items are harness properties, not cell defects: metal density is a check on a *filled* die, and the one differing count (`M1.j`, min 35 % Metal1) reads 35.0 % over the padded array's bounding box against 37.1 % over the actual cell area (control: 45.2 %).

The library has also carried a real block: two variants of the `spi_slave` IP placed and routed through LibreLane/OpenROAD at 100 MHz pass the full IHP signoff deck — zero geometry violations (section 6).

The device transform

	thin oxide	thick oxide
device model	`sg13_lv_nmos` / `sg13_lv_pmos`	`sg13_hv_nmos` / `sg13_hv_pmos`
gate length	130 nm	**450 nm** (`Gat.a3` minimum)
NMOS width	—	unchanged
PMOS width	—	**× 2.40**, snapped per finger to the 5 nm grid
`as/ad/ps/pd`	—	recomputed from device geometry
supply	1.2 V	3.3 V

Deliberate long-channel devices (decap 1.0 µm, `sighold` 700 nm, `dlygate4sd3_1` 500 nm) keep their lengths; 914 devices moved to 450 nm. The antenna cell is unchanged apart from its name (junction diodes).

Why × 2.40. The thin-oxide library is sized for a centred switching threshold ($V_m/V_{DD} = 0.5046$ at $W_p/W_n = 1.5135$), not drive match. At 3.3 V / 450 nm the PMOS is far weaker (219 vs 533 µA/µm) and the same V_m needs $W_p/W_n = 3.63$, i.e. $K_p = 3.63/1.5135 = 2.40$ on every PMOS with every NMOS untouched — preserving each cell's internal stack ratios. Across the library's 0.3–17.9 µm width range the required ratio stays within ±5 % of 2.40 (`work/vm_sweep.py`, `work/ratio_check.py`).

Parasitics follow the vendor deck's geometry formulas; before any generation they were validated by recomputing all 924 thin-oxide devices (worst relative error 3.75×10^{-4}, the vendor's four printed digits). The generator refuses to run if this validation fails.

Cost, measured on `inv_1` (`work/fo4.py`): input capacitance 2.67 → 5.87 fF (2.20×), FO4 delay 53.6 → 142.4 ps (2.66×). The thick-oxide NMOS delivers *more* current per micron at 3.3 V than the thin-oxide at 1.2 V (533 vs 391 µA/µm) — the loss is capacitance, not drive.

The views

All netlist views are re-emitted from one internal model by `work/gen_hv_lib.py`, so a device cannot silently diverge between views.

- **`spice/`** — 84 subckts, 924 PSP103 devices (OSDI/ngspice); widths synchronised to the drawn layout.
- **`cdl/`** — the same subcircuits as `M`-cards with `*.PININFO`, the LVS reference; compared field-by-field against SPICE by `verify_sch.py`.
- **`verilog/`** — 84 modules; deliberately no copy of `sg13g2_udp.v` (the `ihp_*` UDPs are shared — a second copy would collide).
- **`sym/`, `sch/`** — 84 symbols (thin-oxide drawn geometry, thick-oxide netlist prefix, `$::SG13G2_HV_SCH` resolution) and 84 schematics plus a generated gallery sheet; three-view consistency proven on all 924 devices.
- **`gds/`** — 68 cells: 66 by 1-D retarget (section 4), tie cells built by `work/gen_tie_cells.py`. Uniform 7.140 µm row height (17 tracks), every width a 0.48 µm site multiple, all 5 758 contact/via cuts exactly 0.16 × 0.16 µm, every rail tap on the site-centred 0.48 µm grid (`work/fix_rail_contacts.py`, section 6).
- **`lef/`** — 68 macros + `CoreSiteHV`, generated from the GDS: pins from the pin datatypes labelled by contained text, OBS = drawn metal minus pin geometry, antenna values recomputed from the netlist (the thin-oxide numbers are wrong for 3.5× longer gates). Parsed back with klayout and pin sets verified against the CDL.
- **`lib/`** — Liberty NLDM at 3.3 V / 25 °C typical (section 7).
- **`klayout/`** — a cell-library registration macro: with the repository's `klayout/` directory on `KLAYOUT_PATH`, all 68 drawn cells appear in KLayout's Instance dialog as library `sg13g2_stdcell_hv`.

Layout generation

Hand-drawn 2-D layout cannot be regenerated by placement, but it can be **1-D retargeted**: monotone piecewise-linear coordinate maps applied to every vertex preserve topology, so connectivity survives by construction (`work/layout_retarget.py`). The engineering is in the breakpoints and the exceptions:

- **x-map**: gates widen 0.13 → 0.45 µm about their centres; contacted poly pitch 0.48 → 0.80 µm. One accepted consequence: `dlygate4sd2_1` staggers two gates at overlapping x, so its PMOS lands at a legal 0.625 µm, carried consistently into every netlist view.
- **y-map**: PMOS band × 2.40 plus three channel inserts for the thick-oxide clearances (`NW.d1.dig` 0.215 µm, `pSD.j1` 0.100 µm, `pSD.i1` rail insert 0.430 µm), derived library-wide so every cell keeps one row height.
- **PMOS diffusion** is re-emitted per connected slab piece with channel slabs frozen — the only scheme that keeps exact 5 nm-grid device widths without merging, collapsing or renaming devices (each alternative was tried and caught by LVS or `Act.*` rules).
- **Cut layers are translated, never scaled**; contacts are re-tiled inside the new Activ at 0.36 µm pitch (`Cnt.b1` array spacing), grouped by polygon.
- **ThickGateOx** is drawn on the cell boundary grown 0.27 µm in x and 0.42 µm in y — the y-margin proven by the N-row edge experiment (the TGO.a count stays 2 for any N and moves with the outer edge).
- **Five Metal1 re-routes** (`M1E_EDITS` in the generator) resolve the `M1.e` wide-metal gaps the retarget manufactures; each edit is verified in code (polygon count, `M1.a/b/c1`, 5 nm grid).
- **Netlist ↔ layout sync**: the drawn width is authoritative; `work/sync_netlist_widths.py` brings SPICE and CDL to the drawn geometry per *finger* and recomputes parasitics.
- **Site and tracks**: `TRACK_PAD` stretches the mid-cell dead zone to make exactly 17 horizontal 0.42 µm tracks; `pad_to_site` pads each width to the 0.48 µm site (13.75 µm total across the library). Mean cell area is **2.87×** the thin-oxide library (median 2.83, range 1.89–4.21).

Functional verification

Suite	Coverage	Result
`verify_logic.py`	60 combinational cells, 452 vectors, both libraries in one deck	**PASS**
`verify_seq.py`	16 stateful cells, 400 clocked samples	**PASS**
`verify_sch.py`	84 schematics + CDL vs SPICE, 924 devices field-by-field	**PASS**

The thin-oxide library is the golden reference — the transform changed devices, never logic. The 12 high-impedance states of the tri-state cells are exercised; illegal set/reset combinations are excluded rather than counted as passes; both suites were re-run on the final, shipped netlists.

Physical sign-off

Methodology. Cells are checked in abutted context, never standalone. Two harnesses: `work/make_drc_top.py` (every cell in its own column, mirrored second row at the padded bbox pitch) for cell-internal rules, and `work/make_shared_rail_rows.py` — rows at the true 7.14 µm pitch, orientations N/S/FS, rotated cell order per row, cells advanced by LEF width — for everything only a placed block exhibits. Measurement rules, each bought with a wrong conclusion: one fixed runner invocation (counts from different flag sets are not comparable), diagnosis from flat `klayout.db` rule replication (deep-mode marker coordinates mis-attribute cells), and the IHP thin-oxide library run through the identical harness as control. Pass/fail is parsed from the report database, never the exit code.

DRC. 781 raw violations after the first retarget were driven to zero cell-rule items (density-only residue, same class as the control). One defect class survived the original harness because it could not express it: the retarget left each cell's **rail-tap contacts** at cell-specific x, and in a placed block — where *different* cells share every rail — partially overlapping taps merged into 0.19–0.32 µm bars: ~19 000 `Cnt/CntB` markers on the first `spi_slave` signoff. `work/fix_rail_contacts.py` re-tiles every rail tap onto the site-centred 0.48 µm grid, which both the FS row flip and placement's x-mirror preserve, with in-code guards for implant polarity, enclosures, gate clearance, contact spacing and tap continuity (1 134 → 1 810 rail contacts). Full analysis with figures: the companion report *The Shared-Rail Contact Clash in sg13g2_stdcell_hv* (`shared_rail_contact_fix.pdf`, distributed alongside the library rather than inside it).

LVS. All 68 drawn cells match, re-run in full after the re-tiling. The four device-less `fill_*` cells need the runner's own `--ignore_top_ports_mismatch` relaxation — IHP's own fill cells fail the identical flow the same way. LVS caught five defect classes dimensional checks could not (merged channels, collapsed notch devices, mis-paired fingers, a dropped substrate tie, stale diode geometry), which is why the per-cell sweep runs after every layout change.

Block-level validation. The `spi_slave` IP, in two variants, was synthesised, placed and routed with this library through LibreLane/OpenROAD at 100 MHz on ~354 × 400 µm dies. Both close timing with zero setup/hold violations at the characterised corner, pass routing DRC, antenna and Netgen LVS in the flow, pass gate-level simulation of the routed netlist (all four SPI modes on the all-modes variant), and pass the **full IHP signoff deck** with zero geometry violations — only the 8 chip-level density/filler markers a filled die resolves at tapeout. The flow settings this takes (flip-flop mapping onto `sdfbbp_1`, excluded cells, Metal1 routing, rails-only fillers) are documented in the `spi-slave-ihp` repository.

Liberty characterisation

`lib/sg13g2_stdcell_hv_typ_3p30V_25C.lib` is produced with **CharLib** against the shipped, layout-synchronised SPICE netlist on the PSP103/OSDI models — 25 839 combinational simulations plus a 2 962-task sequential run: 600 delay/slew tables over 66 cells (52 combinational, 9 flip-flops, 5 latches), none empty. Grids are the thin-oxide grids rescaled by the measured 2.66× delay / 2.20× capacitance ratios; boolean functions are translated from the thin-oxide Liberty and checked by truth-table equivalence. Stock CharLib needed five committed tool workarounds (OSDI shim, `ngspice-shared` backend, case-insensitive supply lookup, Liberty syntax post-pass) and — for sequential cells, which stock CharLib cannot characterise at all — custom clk→Q and setup/hold-bisection procedures with a c2q-degradation pass criterion (`work/seq_delay_procedure.py`).

The shipped file is verified **as data** by `work/verify_lib.py`: structure (600/600 tables populated), cross-view (all pins exist in the CDL), physical (every `area` equals the drawn boundary to 1 nm²), monotonicity along the load axis (1 931/1 932 series, one documented pessimistic waiver: `xnor2_1` glitch latching), input capacitance against an independent both-rails measurement (6.44 vs 5.87 fF, 9.7 % — this check caught a 6.8–7.5× Miller defect in CharLib's default procedure), and sequential arcs (14/14 cells: delay + setup + hold present, nothing pinned at search bounds). A hand-written transient at a table corner agrees to 0.3 %.

An independent characteriser, **lctime**, was run over eight cells on the same grids and models: in the region STA exercises (real loads, slews < 1 ns) the two agree to a median 2.9 % on delays and 0.0 % on transitions; every disagreement region traces to a stimulus convention, and a direct measurement sides with the shipped table (+0.15 % vs −19 % at the probe point). lctime's input capacitance (+52 %) also corroborates validating Cin against a direct measurement, not a characteriser.

Excluded from the Liberty, by configuration and with stated reasons: the device-less cells, `tiehi`/`tielo`, the statetable clock gates, and the 6 tri-state cells (CharLib cannot express high-impedance outputs).

Verification summary

Check	Method	Scope	Result
Parasitic formulas	recompute vendor `as`/`ad`/`ps`/`pd`	all 924 thin-oxide devices	PASS, worst error 3.75×10^{-4}
Combinational logic	ngspice vs thin-oxide golden	60 cells, 452 vectors, 12 high-Z states	**PASS**
Sequential logic	ngspice, clocked walk from reset	16 stateful cells, 400 samples	**PASS**
Three-view consistency	symbols vs SPICE vs CDL	84 cells, 924 devices	**PASS**

Check	Method	Scope	Result
DRC, padded array	PDK deck, fixed invocation	68 drawn cells	**0 cell rules**; 7 density items
DRC, shared-rail array	same deck, true-pitch mixed/mirrored rows	68 cells × 4 rows	**0 cell rules**; density only
DRC control	identical harness on IHP `sg13g2_stdcell`	84 cells	0 cell rules, 6 density
LVS	PDK deck, per cell, after the rail re-tiling	68 cells vs CDL	**68/68**
LEF	klayout parse-back + pin sets vs CDL	68 macros	PASS, on-grid
P&R block validation	LibreLane `spi_slave` ×2 + full signoff deck	~354 × 400 µm, 100 MHz	all clean; signoff **0 geometry violations**, 8 density markers
Liberty structure/ views/areas	`verify_lib.py` 1–3	66 cells	PASS, areas exact to 1 nm²
Liberty monotonicity	`verify_lib.py` 4	1 932 delay series	1 931 + 1 documented waiver
Liberty Cin	vs both-rails reference	`inv_1`	6.44 vs 5.87 fF (9.7 %)
Liberty delay point-check	hand-written transient	table corner	0.3 %
Independent characteriser	lctime, 8 cells, 3 132 points	STA region	median 2.9 % / 0.0 %
Sequential arcs	`verify_lib.py` 6	14 flip-flops/latches	**14/14 clean**
Site geometry	boundary scan	all 68 cells	7.140 µm, widths on 0.48 µm site

Known limitations

- **16 cells have no layout** — eight D-flip-flop variants run NMOS Activ to the channel cut, eight others run PMOS Activ into the VDD rail; neither scales at a shared row height. All have full netlist, schematic, symbol and Verilog views. (`tiehi`/`tielo`, once on this list, were rebuilt by `work/gen_tie_cells.py`.)
- **P&R needs non-standard flow settings** — flip-flop mapping onto `sdfbbp_1`, an excluded-cell list, Metal1 routing for the off-grid pins, rails-only fillers (see the `spi-slave-ihp` flow configuration).
- **The 6 tri-state cells carry no Liberty timing**; they are simulation-verified.

- **One corner** (typical, 3.3 V, 25 °C); sequential leakage is a single-settled-state number; the delay cells' ratios shift with the 450 nm minimum; the decaps store less per unit area.
- **Not silicon-proven, and not independently reviewed.**

Reproducing the library

```
cd /foss/designs/sg13g2_stdcell_hv/work

python3 gen_hv_lib.py            # netlists, schematics, symbols, Verilog
python3 gen_gallery.py           # gallery sheet
python3 layout_retarget.py       # gds/ (66 cells; prints the 16 skips)
python3 gen_tie_cells.py         # + tiehi / tielo
python3 fix_rail_contacts.py     # rail taps onto the site-centred grid
python3 sync_netlist_widths.py   # SPICE + CDL follow the drawn geometry
python3 gen_lef.py               # lef/ (CoreSiteHV + 68 macros)
python3 make_drc_top.py          # padded two-row DRC context
python3 make_shared_rail_rows.py # shared-rail mixed-row DRC context

python3 verify_logic.py; python3 verify_seq.py; python3 verify_sch.py
./run_lvs.sh                     # per-cell LVS, 68/68
# DRC: PDK run_drc.py on drc/drc_top.gds and drc/shared_rail.gds

./run_charlib.sh ../lib/sg13g2_stdcell_hv_typ_3p30V_25C.lib
# sequential cells via charlib_patched.py, then:
python3 merge_lib.py ../lib/sg13g2_stdcell_hv_typ_3p30V_25C.lib <seq.lib>
python3 seq_leakage.py
python3 verify_lib.py            # gates the shipped Liberty as data
python3 lctime_compare.py        # independent characteriser cross-check
```

Every number in this report is reproducible from these scripts; the `work/` directory and the `README.md` are the provenance of record.